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 Studierichting: Informatica
 Oefeningenreeks 3A nr. 40.7

$$\begin{aligned}
 & \frac{df}{dx} \left(\frac{\sqrt{2x^3 - 2x + 1}}{\sqrt[3]{x + \frac{1}{x}}} \right) \\
 &= \frac{(\sqrt{2x^3 - 2x + 1})' \sqrt[3]{x + \frac{1}{x}} - \left(\sqrt[3]{x + \frac{1}{x}} \right)' (\sqrt{2x^3 - 2x + 1})}{\sqrt[3]{\left(x + \frac{1}{x}\right)^2}} \\
 &= \frac{\frac{1}{2} \sqrt[3]{x + \frac{1}{x}} (2x^3 - 2x + 1)^{-\frac{1}{2}} (6x^2 - 2) - \left(\frac{1}{3} \left(x + \frac{1}{x}\right)^{-\frac{2}{3}} \left(1 - \frac{1}{x^2}\right) (2x^3 - 2x + 1)^{\frac{1}{2}} \right)}{\left(x + \frac{1}{x}\right)^{\frac{2}{3}}} \\
 &= \frac{\left(x + \frac{1}{x}\right)^{-\frac{2}{3}} (2x^3 - 2x + 1)^{-\frac{1}{2}} \left(\frac{1}{2} (6x^2 - 2) \left(x + \frac{1}{x}\right) - \left(\frac{1}{3} \left(1 - \frac{1}{x^2}\right) (2x^3 - 2x + 1) \right) \right)}{\left(x + \frac{1}{x}\right)^{\frac{2}{3}}} \\
 &= \left(x + \frac{1}{x}\right)^{-\frac{4}{3}} (2x^3 - 2x + 1)^{-\frac{1}{2}} \left((3x^2 - 1) \left(x + \frac{1}{x}\right) - \frac{1}{3} \left(1 - \frac{1}{x^2}\right) (2x^3 - 2x + 1) \right) \\
 &= \left(x + \frac{1}{x}\right)^{-\frac{4}{3}} (2x^3 - 2x + 1)^{-\frac{1}{2}} \left(\left(3x^3 + 2x - \frac{1}{x}\right) - \frac{1}{3} \left(2x^3 - 4x + \frac{2}{x} - 1 - \frac{1}{x^2}\right) \right) \\
 &= \left(x + \frac{1}{x}\right)^{-\frac{4}{3}} (2x^3 - 2x + 1)^{-\frac{1}{2}} \left(\frac{9x^3}{3} + \frac{6x}{3} - \frac{3}{3x} - \frac{2x^3}{3} + \frac{4x}{3} - \frac{2}{3x} - \frac{1}{3} + \frac{1}{3x^2} \right) \\
 &= \left(x + \frac{1}{x}\right)^{-\frac{4}{3}} (2x^3 - 2x + 1)^{-\frac{1}{2}} \left(\frac{7x^3}{3} + \frac{10x}{3} - \frac{5}{3x} - \frac{1}{3} + \frac{1}{3x^2} \right)
 \end{aligned}$$

References